

Class17

Kate Ruiz (PID: A17671200)

Core Unix commands

Most unix commands have super short names, which makes them quick to type but annoying to learn. Major file system related commands include: pwd: Print working directory ls: list files and directories cd: change directory mkdir: make a new directory rm: remove files and directories (delete) super dangerous cp: copy files (source > destination) mv: move files or directories (basically rename) nano: a wee text command editor (very basic but always available)

AW EC2 Instance

Connect to my instance with:

```
ssh -i ~/Downloads/bimm143_kate.pem ubuntu@ec2-34-219-54-212.us-west-2.compute.amazonaws.com
```

Secure Copy files between machines, in this case from our instance to our laptop

```
scp -i ~/Downloads/bimm143_kate.pem ubuntu@ec2-34-219-54-212.us-west-2.compute.amazonaws.com:/home/ubuntu/
```

PCA

```
library(tximport)

# setup the folder and file-names to read
folders <- dir(pattern="SRR21568*")
samples <- sub("_quant", "", folders)
files <- file.path( folders, "abundance.h5" )
names(files) <- samples

txi.kallisto <- tximport(files, type = "kallisto", txOut = TRUE)
```

1 2 3 4

```
head(txi.kallisto$counts)
```

	SRR2156848	SRR2156849	SRR2156850	SRR2156851
ENST00000539570	0	0	0.00000	0
ENST00000576455	0	0	2.62037	0
ENST00000510508	0	0	0.00000	0
ENST00000474471	0	1	1.00000	0
ENST00000381700	0	0	0.00000	0
ENST00000445946	0	0	0.00000	0

```
getwd()
```

```
[1] "/Users/kateruiz/Desktop/BIMM 143/Class17"
```

```
list.files()
```

```
[1] "Class17.pdf"      "Class17.qmd"      "Class17.rmarkdown"
[4] "Class17.Rproj"    "SRR2156848_quant" "SRR2156849_quant"
[7] "SRR2156850_quant" "SRR2156851_quant"
```

```
folders <- dir(pattern = "SRR21568")
folders
```

```
[1] "SRR2156848_quant" "SRR2156849_quant" "SRR2156850_quant" "SRR2156851_quant"
```

```
samples <- sub("_quant", "", folders)
samples
```

```
[1] "SRR2156848" "SRR2156849" "SRR2156850" "SRR2156851"
```

```
files <- file.path(folders, "abundance.h5")
files
```

```
[1] "SRR2156848_quant/abundance.h5" "SRR2156849_quant/abundance.h5"
[3] "SRR2156850_quant/abundance.h5" "SRR2156851_quant/abundance.h5"
```

```
names(files) <- samples
files
```

```
                SRR2156848                SRR2156849
"SRR2156848_quant/abundance.h5" "SRR2156849_quant/abundance.h5"
                SRR2156850                SRR2156851
"SRR2156850_quant/abundance.h5" "SRR2156851_quant/abundance.h5"
```

```
txi.kallisto <- tximport(files, type = "kallisto", txOut = TRUE)
```

```
1 2 3 4
```

```
if (!requireNamespace("BiocManager", quietly = TRUE))
  install.packages("BiocManager")
```

```
BiocManager::install("rhdf5")
```

Bioconductor version 3.22 (BiocManager 1.30.27), R 4.5.2 (2025-10-31)

Warning: package(s) not installed when version(s) same as or greater than current; use
`force = TRUE` to re-install: 'rhdf5'

```
colSums(txi.kallisto$counts)
```

```
SRR2156848 SRR2156849 SRR2156850 SRR2156851
      2563611      2600800      2372309      2111474
```

```
sum(rowSums(txi.kallisto$counts)>0)
```

```
[1] 94561
```

```
to.keep <- rowSums(txi.kallisto$counts) > 0
kset.nonzero <- txi.kallisto$counts[to.keep,]
```

```
keep2 <- apply(kset.nonzero,1,sd)>0
x <- kset.nonzero[keep2,]
```

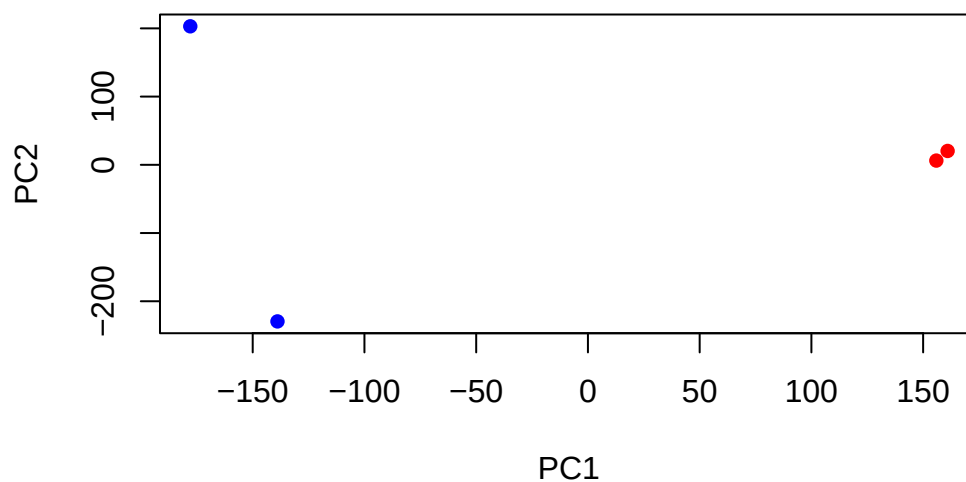
```
pca <- prcomp(t(x), scale=TRUE)
```

```
summary(pca)
```

Importance of components:

	PC1	PC2	PC3	PC4
Standard deviation	183.6379	177.3605	171.3020	1e+00
Proportion of Variance	0.3568	0.3328	0.3104	1e-05
Cumulative Proportion	0.3568	0.6895	1.0000	1e+00

```
plot(pca$x[,1], pca$x[,2],  
     col=c("blue","blue","red","red"),  
     xlab="PC1", ylab="PC2", pch=16)
```



```
names(pca)
```

```
[1] "sdev"      "rotation" "center"    "scale"     "x"
```

```
head(pca$x)
```

	PC1	PC2	PC3	PC4
SRR2156848	-177.9368	203.031882	-4.507483	0.8660196
SRR2156849	-138.9188	-229.558755	8.656814	0.8659919
SRR2156850	155.8981	6.206921	-211.755452	0.8660168
SRR2156851	160.9486	20.312009	207.599341	0.8660462

```
df <- as.data.frame(pca$x)  
head(df)
```

	PC1	PC2	PC3	PC4
SRR2156848	-177.9368	203.031882	-4.507483	0.8660196
SRR2156849	-138.9188	-229.558755	8.656814	0.8659919
SRR2156850	155.8981	6.206921	-211.755452	0.8660168
SRR2156851	160.9486	20.312009	207.599341	0.8660462

```
rownames(pca$x)
```

```
[1] "SRR2156848" "SRR2156849" "SRR2156850" "SRR2156851"
```

```
colnames(df)
```

```
[1] "PC1" "PC2" "PC3" "PC4"
```

```
head(df)
```

	PC1	PC2	PC3	PC4
SRR2156848	-177.9368	203.031882	-4.507483	0.8660196
SRR2156849	-138.9188	-229.558755	8.656814	0.8659919
SRR2156850	155.8981	6.206921	-211.755452	0.8660168
SRR2156851	160.9486	20.312009	207.599341	0.8660462

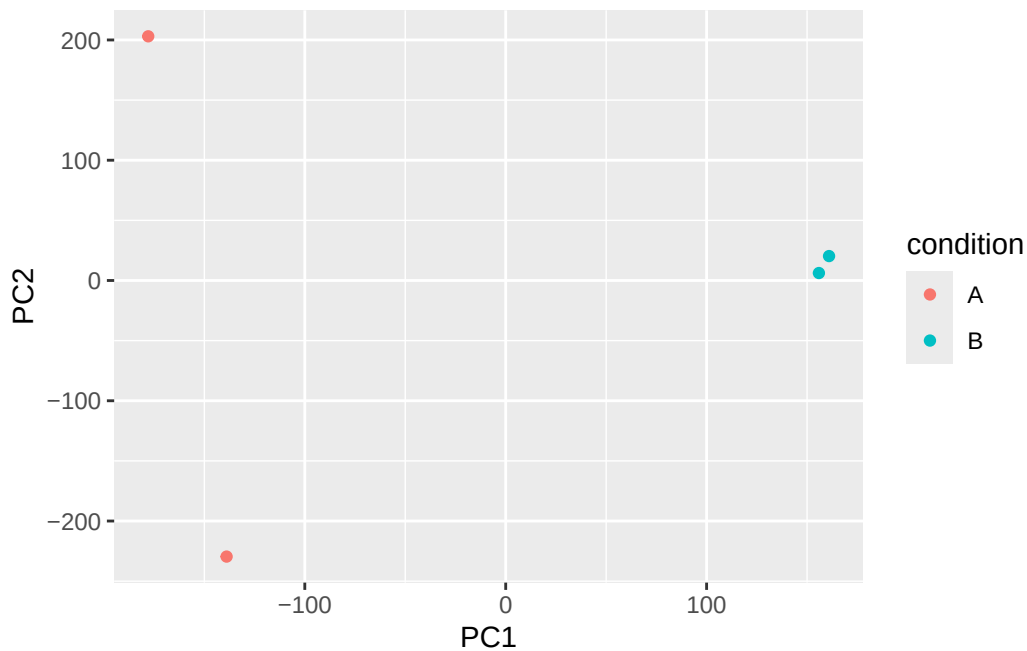
```
df$condition <- c("A", "A", "B", "B")  
head(df)
```

	PC1	PC2	PC3	PC4	condition
SRR2156848	-177.9368	203.031882	-4.507483	0.8660196	A
SRR2156849	-138.9188	-229.558755	8.656814	0.8659919	A
SRR2156850	155.8981	6.206921	-211.755452	0.8660168	B
SRR2156851	160.9486	20.312009	207.599341	0.8660462	B

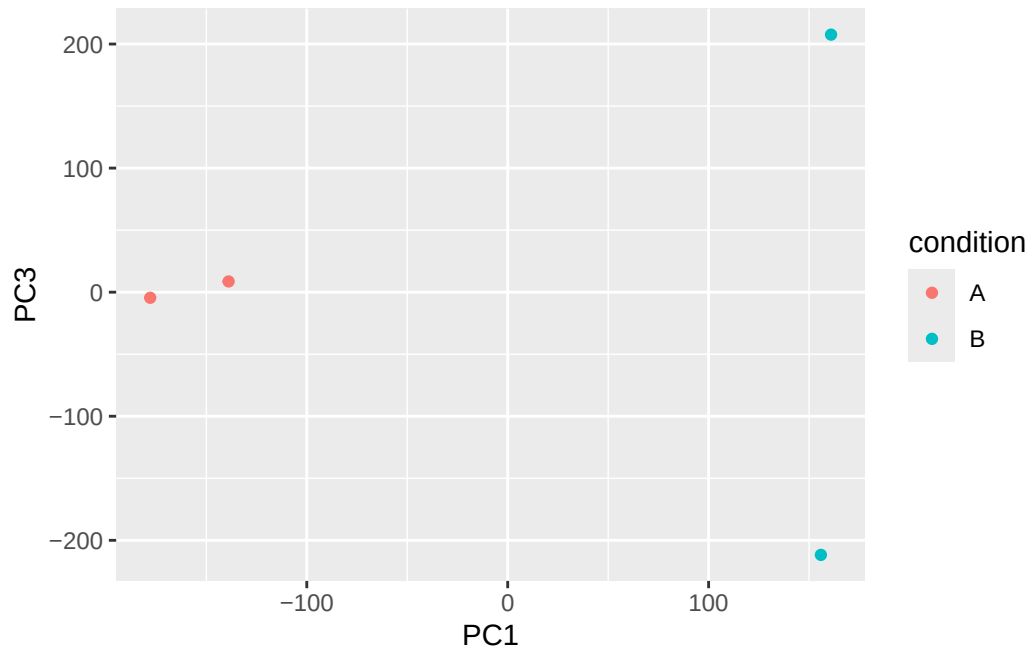
```
head(df)
```

	PC1	PC2	PC3	PC4	condition
SRR2156848	-177.9368	203.031882	-4.507483	0.8660196	A
SRR2156849	-138.9188	-229.558755	8.656814	0.8659919	A
SRR2156850	155.8981	6.206921	-211.755452	0.8660168	B
SRR2156851	160.9486	20.312009	207.599341	0.8660462	B

```
library(ggplot2)
ggplot(df, aes(x = PC1 , y = PC2 , color = condition)) +
  geom_point()
```



```
ggplot(df, aes(x = PC1 , y = PC3 , color = condition)) +
  geom_point()
```



```
ggplot(df, aes(x = PC2 , y = PC3 , color = condition)) +  
  geom_point()
```

